

Data for Policy Analysis and Management

CLASS 1: INTRODUCTION,
SOCIAL POLICY AND
USING R

Instructor: Aida Pacheco-Applegate
Summer 2026

Agenda

1. Introductions
2. Syllabus
3. Social Policy
4. Empirical Research
5. Lab Session: Introduction to R

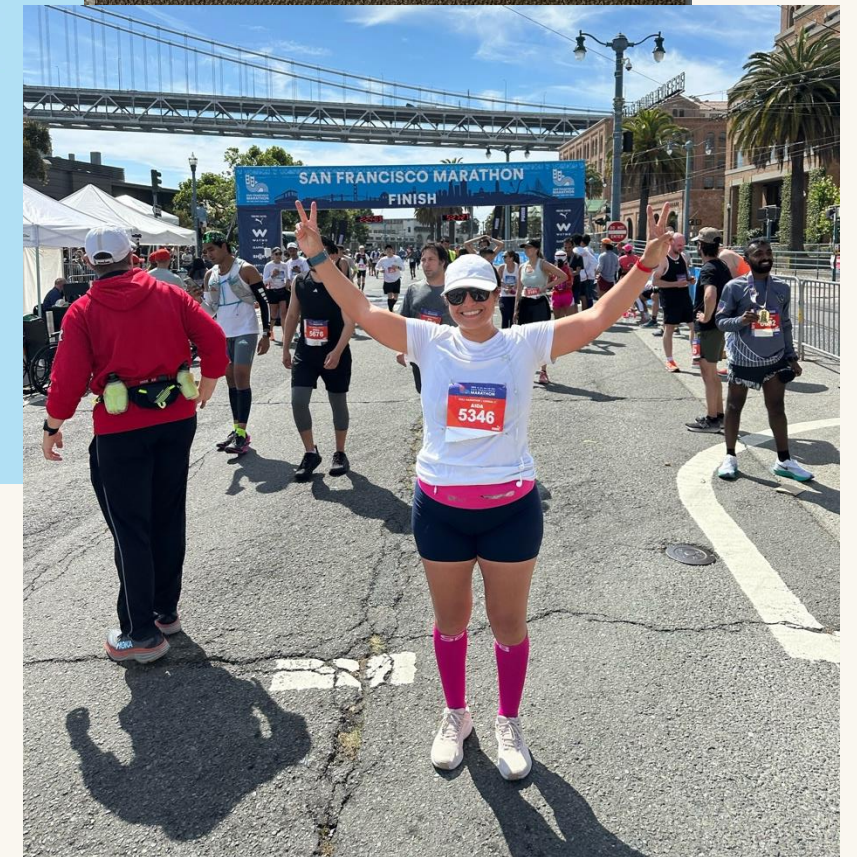
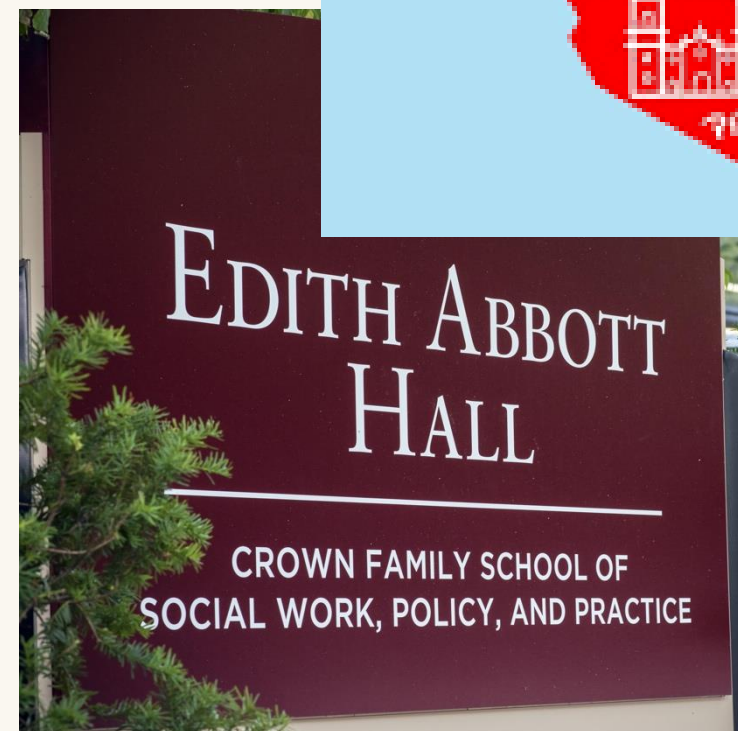
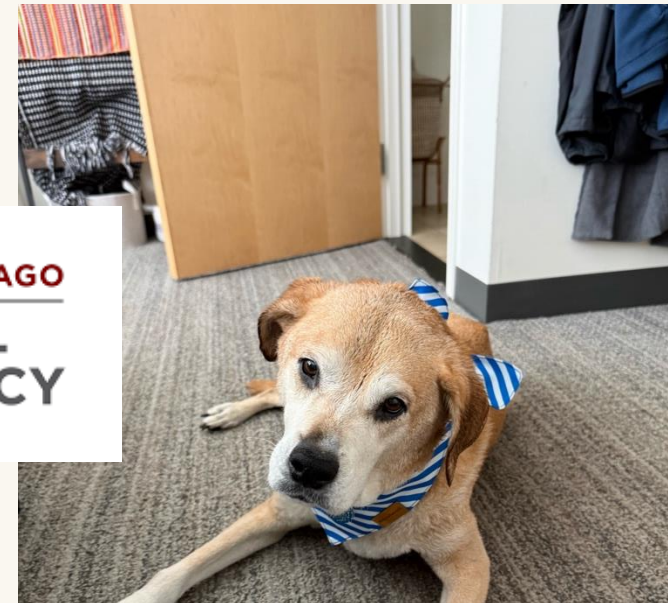
Introduction

Share with the class:

1. Your name and pronouns
2. Where are you from/where did you grow up?
3. What's one thing you're hoping to get out of this class?
4. Something fun/interesting about yourself



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Let's review the syllabus

Social Policy

What it is?

Social policy refers to the actions, rules, and programs created by governments and organizations to address social issues and improve well-being.

Key Areas

- Health care
- Education
- Housing
- Employment
- Income support
- Child welfare

Why It Matters

- Improves well-being
- Reduces inequality
- Expands access to services
- Supports economic stability

Social Policy and Social Work

- Social policy shapes how social work is practiced
- It affects the resources and opportunities available to people and communities
- Many advocates and professionals argue that social workers should engage in public policy and politics
- Social workers may influence policy through:
 - Policy practice
 - Professional organizations
 - Academic work



Empirical Research

What is empirical research?

- Uses real-world observations (data) to answer research questions.
- Can use surveys, interviews, administrative data, or experiments.
- Focuses on evidence rather than theory or intuition alone.

Quantitative research

- Uses numbers and measurable data.
- Focuses on patterns, relationships, and statistical analysis

Qualitative research

Uses interviews, observations, and text
Focuses on meaning, experiences, and context

Statistical Science Role

- Design studies (plan how to collect data)
- Describe data (summarize what we observe)
- Make inferences (draw conclusions beyond the data)

Statistical Science in Empirical Research

1

Design

- **Definition:** How we plan to collect data
- Includes decisions about:
 - Who is studied
 - What is measured
 - How data is collected
- **Goal:** Ensure data can actually answer the research question

2

Description

- **Definition:** Organizing and summarizing data
- **Uses:**
 - Tables, graphs, averages, percentages, etc.
 - Shows patterns in the observed data.
- **Goal:** Make data easy to understand

3

Inference

- **Definition:** Using data to make general conclusions or predictions.
- Moves from sample → population
- Deals with uncertainty in a structured way
- **Goal:** Answer questions beyond what is directly observed

Some notes about study design...

- Data alone does not automatically answer the research question
- Poor design can lead to wrong or misleading conclusions
- Helps distinguish correlation vs. causation
- Ensures we are measuring the right thing in the right way
- Good design is what makes valid inference possible



Next Class

- Data, measurement and variables
- In-class activities
- Henning-Smith, et al. (2024). [paper](#)

Questions?

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15-minute break